# interneurOn: A Collapsing-Bound Decision Neuron for Neuromorphic Early Exit

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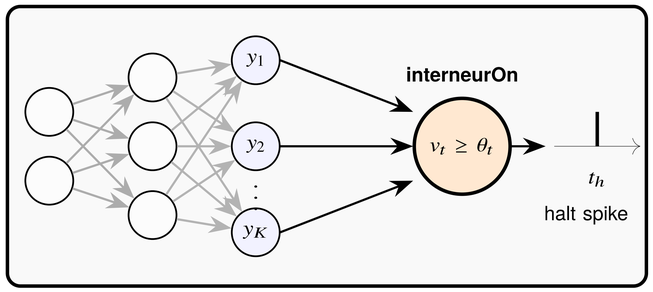
## Motivation

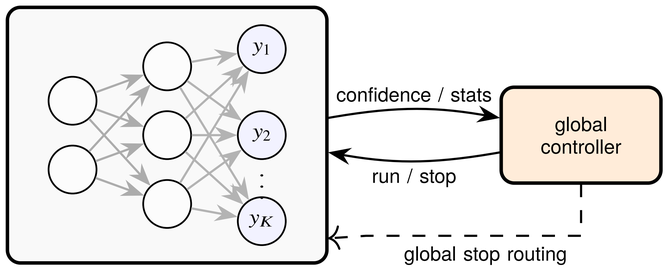
Digital neuromorphic systems often run inference to a fixed horizon because they lack a reliable on-chip stop signal, wasting computation after sufficient evidence has already emerged. Existing spiking early-exit methods often determine stopping externally from output confidence, introducing coordination and routing overhead. We present interneurOn, a single spike-domain neuron that generates a local stop signal from channel activity.

**(a) Spike-domain *interneurOn* attached to the output head**




SNN + interneurOn

**(b) Global early-exit controller**




SNN

**(a) interneurOn**
controls network activity locally

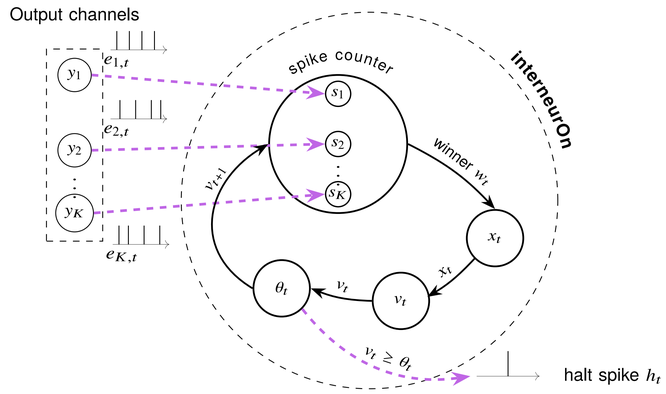
**(b) Traditional early exit**
requires an external controller

self-termination
interneuron
early-exit
evidence
accumulator
On-chip

We evaluate **interneurOn**
on classification and graph propagation

no external aggregation
no extra coordination overhead
local to the evidence stream
hardware-friendly design
bounded per-tick computation

## Spike-Domain Signal Path



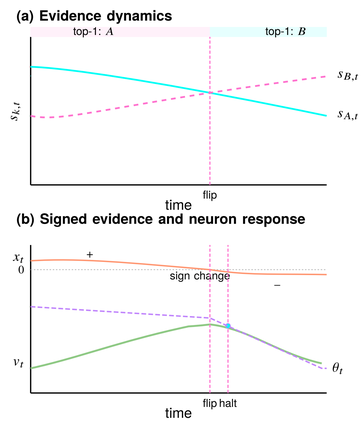

**System model:** evidence $e_{k,t} \longrightarrow$ state $s_{k,t} \longrightarrow$ margin $x_t \longrightarrow$ decision $v_t \longrightarrow v_t \geq \theta_t \longrightarrow$ halt spike

## Decision Dynamics and Per-Tick Update

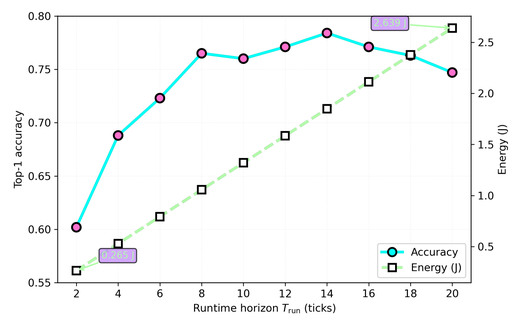
At each tick, interneurOn performs four local operations:

1. **Evidence update**
$$s_{k,t} = (1-\eta)s_{k,t-1} + e_{k,t}$$
2. **Locked top-2 margin**
$$x_t = s_{c^\star,t} - \max_{j \neq c^\star} s_{j,t}$$
3. **Leaky integration**
$$v_t = (1-\lambda)v_{t-1} + g\,x_t$$
4. **Collapsing bound**
$$\theta_t = \max(\theta_{\min},\ b_0 - \beta t)$$

A halt spike is emitted once accumulated evidence remains above the bound.



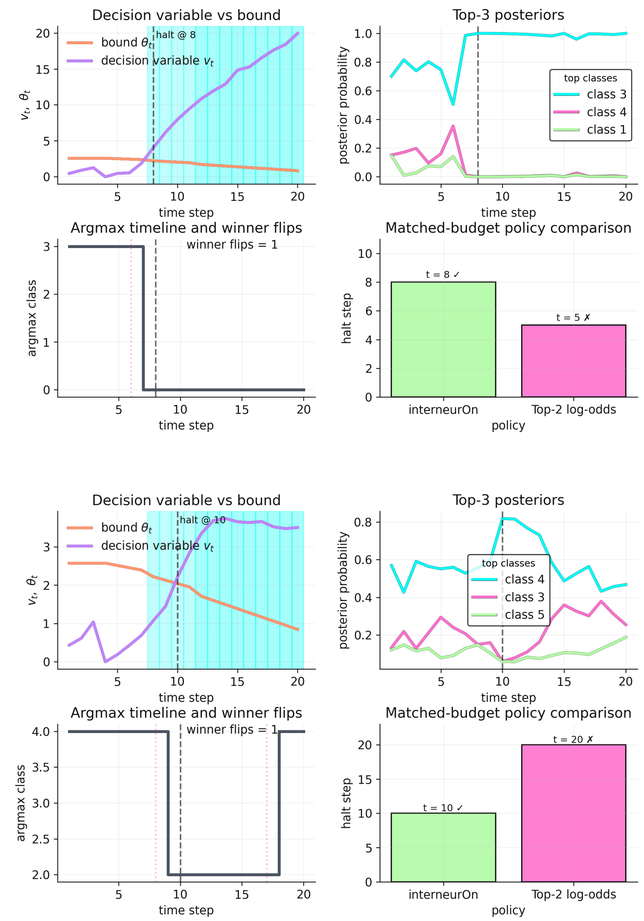

## Measured Energy–Accuracy Trade-off




**Result:** On INSHEP radar classification dataset($N = 375$), each point shows accuracy at decision versus measured SpiNNaker-1 execution energy for a clamped runtime horizon $T_{\text{run}} \in \{2, 4, \ldots, 20\}$.

## Per-Stream Behaviour

Radar test streams show interneurOn halting at a stable positive margin.




## SpiNNaker-1 Implementation

**On-chip evidence interface**

Output spikes from the 6 class channels are accumulated into per-tick counts $e_{k,t}$. These counts update the local evidence state.

**Setting:** INSHEP radar, 6 classes, 1 ms tick.

| Item | Value |
|---|---|
| Parameters / neuron | 40 B |
| State / neuron | 96 B |
| Cycles / tick / core | 33 974 |
| Cycles / tick / neuron | 531 |
| Core utilisation / tick | 17.0% |

**Per-tick work, small on-core state**

## Stored State and Stabilization

| State | Role |
|---|---|
| $\mathbf{s}_{1:K}$ | evidence state |
| $\mathbf{v}$ | decision variable |
| $\mathbf{c}^\star$ | locked hypothesis |
| $\mathbf{w}_{\text{prev}}$ | previous winner |
| $\mathbf{d}$ | dwell counter |
| $\ell$ | lock-out counter |
| `halted` | halt flag |

**Control state.** Beyond the evidence state $\mathbf{s}_{1:K}$ and decision variable $v$, the neuron stores the locked hypothesis $c^\star$, the previous winner $w_{\text{prev}}$, and the halt flag `halted`. These variables keep track of winner identity and prevent further updates after commitment.

**Stabilization logic.** The dwell counter $d$ enforces persistence before halting, the lock-out counter $\ell$ briefly pauses updates after winner flips, and negative reset clears the decision state when strong opposing evidence drives the accumulator in the wrong direction.

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